

11.6 Rand Index

Q1: A B C D

Method 1 $C_1 = \{A, B\}$ Method 2 $C_1 = \{A, C\}$
 $C_2 = \{C, D\}$ $C_2 = \{B, D\}$

Calculate: Rand Index

Solution ① $RI = \frac{N_s + N_d}{C(4, 2)}$

$$C_4^2 = \frac{4 \times 3}{1 \times 2} = 6$$

	Method 1	Method 2	N_s / N_d
A B	✓	X	
A C	X	✓	
A D	X	X	$N_d + 1$
B C	X	X	$N_d + 1$
B D	X	✓	
C D	✓	X	

$$RI = \frac{0 + 2}{6} = \frac{1}{3}$$

Q 2 : $RI = 0$ situation

C_1 A

$C_1 = A B C$

C_2 B

C_3 C

Calculate : Rand Index

Solution ① $RI = \frac{N_s + N_d}{C(4,2)}$

$$C_3^2 - \frac{3 \times 2}{1 \times 2} = 3$$

	Method 1	Method 2	N_s / N_d
A B	X	✓	
A C	X	✓	
A D	X	✓	
B C	X	✓	
B D	X	✓	
C D	X	✓	

$$RI = \frac{0 + 0}{3} = 0$$